

I built a crypto market scanner in Python.

Version 1 didn't work.

Here's what the data showed me, the errors I hit, and how I rebuilt it around evidence instead of assumptions.

[Python](#) | [pandas](#) | [Bybit API](#) | [backtesting](#) | [Telegram automation](#)

A coin jumped +83% in one day.

But the real signal came earlier: a long, quiet, flat base (circled). I wanted a tool that finds these setups BEFORE the move.



Step 1: pull live data for every coin

Error #1 on day one: my network couldn't reach Bybit's API. Fix: auto-fallback to their backup domain.

```
op/Crypto-scanner/accumulation_scanner.py
Could not reach api.bybit.com, trying next...
Connected to Bybit via api.bytick.com
[2026-09-28 11:53] Scanning 218 Bybit spot pairs...
█
```

```
Connected to Bybit via api.bytick.com
[2026-09-28 11:53] Scanning 218 Bybit spot pairs...
50/218 done
100/218 done
150/218 done
200/218 done
```

```
Saved all results to scan_20260928_1202.csv
ACCUMULATION SCANNER - 28 Sep 12:02
```

218

coins scanned

1D + 4H

candles per coin

\$0

data cost

It worked: 218 coins scored, ranked, saved and sent to my phone

1	symbol	score	price	drawdown	base_range	extension	recent_run_3d	volume_ratio	reas
2	GRAM/USDT	90	1.686	0.796619983	0.21478599221789896	0.3120622568093385	0.19829424307036247	2.866213529705912	80%
3	SNX/USDT	85	0.2443	0.953753833396434	0.33668607728957123	0.2932768660667018	0.005763689	1.4677404811831316	95%
4	PLUME/USDT	80	0.01764	0.9288422751109319	0.39893617021276584	0.5638297872340425	0.037647059	1.5477872193814266	93%
5	XDC/USDT	75	0.03478	0.7775645945254541	0.18529746115953016	0.31792345585449033	0.1734143049932524	4.098356793213137	78%
6	JASMY/USDT	75	0.005101	0.9135789919525625	0.5780652749191415	0.49985298441634796	0.1328003553186763	2.989201840751629	91%
7	RESOLV/USDT	70	0.01855	0.9555688622754491	0.35374592833876206	0.2084690553745927	0.012002182	2.6367000943092784	96%
8	YFI/USDT	70	2357	0.8403677839567253	0.3591511936339522	0.25039787798408497	0.014199655765920749	2.8550373032766796	84%
9	XAI/USDT	65	0.009085	0.9952184210526316	0.3861207038316068	0.493997698	0.028878822	1.1681561678272934	100%
10	APE/USDT	65	0.15519	0.9419394664970631	0.32259136212624595	0.288953488	0.026864289022695775	1.0354152324733628	94%
11	SAFE/USDT	65	0.10883	0.9758155555555555	0.3739204476341076	0.3238048899160684	0.011901441190143913	0.8453718409077517	98%
12	TRIA/USDT	65	0.004316	0.91624134	2.5960526315789476	0.4197368421052632	0.13370107696348832	3.3024366780772243	92%
13	MON/USDT	65	0.02717	0.4492195418609365	0.63857868	0.3791878172588834	0.079459674	1.9653807605420472	high
14	SCR/USDT	65	0.02375	0.9842087765957447	0.48244680851063815	0.26329787234042556	0.014090520922288574	2.1533465286875675	98%
15	DYDX/USDT	65	0.13412	0.970386399	0.49670122525918936	0.4045449785317832	0.025539073	1.3554311315896501	97%
16	SAND/USDT	65	0.04342	0.9593049411411861	0.6052227342549921	0.333947773	0.020206766917293173	1.7037816000813597	96%
17	ME/USDT	60	0.07485	0.9948725501613246	0.3410485709275821	0.3124671225670701	0.018090315560391756	1.1458263511024334	99%
18	SOSO/USDT	60	0.3288	0.8074264964273163	0.45191927186387026	0.30114760585674727	0.11193777477172806	0.9495542581066043	81%
19	PARTI/USDT	60	0.02631	0.9402520722152833	0.40312815338042385	0.327447023	0.047789725	1.2637489704040703	94%
20	IO/USDT	60	0.1575	0.9758016193709957	0.4316081330868762	0.4556377079482439	0.021400778210116655	0.7384854109143366	98%

ACCUMULATION SCANNER - 28 Sep 12:02

GRAM/USDT score 90 price 1.686
80% below high; tight 45d base (21% range); higher lows; above trend line; MACD above zero; volume waking up (2.9x)

SNX/USDT score 85 price 0.2443
95% below high; tight 45d base (34% range); higher lows; above trend line; fresh trend flip; volume waking up (1.5x)

PLUME/USDT score 80 price 0.01764
93% below high; tight 45d base (40% range); higher lows; above trend line; volume waking up (1.5x)

XDC/USDT score 75 price 0.03478
78% below high; tight 45d base (19% range); higher lows; above trend line; MACD above zero; volume spike (4.1x), may be late

JASMY/USDT score 75 price 0.005101
91% below high; higher lows; above trend line; fresh trend flip; MACD above zero; volume waking up (3.0x)

RESOLV/USDT score 70 price 0.01855
96% below high; tight 45d base (35% range); higher lows; volume waking up (2.6x)

YFI/USDT score 70 price 2357
84% below high; tight 45d base (36% range); higher lows;

The top picks looked right on the chart

Long base, then a push through the 200-day average (blue line). Exactly the pattern I was hunting.



"Looks right" isn't proof. So I backtested.

First result: low scores beat high scores. But 81% of ALL coins pumped that month, and my test only picked coins that were liquid TODAY. Survivorship bias. The test itself was flawed.

```
PS C:\Users\sodiq\OneDrive\Desktop\Crypto-scanner> python accumulation_scan.py
Backtest: scoring 100 coins as of 30 days ago (futures + tokenomics not included)
50/100 done
```

```
BTC over same period: +6.6%
```

	coins	avg_max_gain %	avg_end_return %	hit_30pct %	avg_max_drop %
bucket					
<50	59	69.4	48.2	81.4	-8.6
50-64	5	38.5	25.2	40.0	-12.8

```
Higher score buckets should show better returns. Saved details to backtest_30days.csv
```

The errors (the honest part)

A 48-minute backtest crashed at the final step over a missing library. Then it crashed AGAIN... because I hadn't saved the file.

```
corr = df["score"].corr(df[f"exc_{H}"], method="spearman")
File "C:\Users\sodiq\AppData\Local\Python\pythoncore-3.14-64\Lib\site-packages\pandas\core\series.py", line 2774, in corr
    result = nanops.nancorr(
        this_values, other_values, method=method, min_periods=min_periods
    )
File "C:\Users\sodiq\AppData\Local\Python\pythoncore-3.14-64\Lib\site-packages\pandas\core\nanops.py", line 87, in _f
    return f(*args, **kwargs)
File "C:\Users\sodiq\AppData\Local\Python\pythoncore-3.14-64\Lib\site-packages\pandas\core\nanops.py", line 1655, in nancorr
    f = get_corr_func(method)
File "C:\Users\sodiq\AppData\Local\Python\pythoncore-3.14-64\Lib\site-packages\pandas\core\nanops.py", line 1670, in get_corr_func
    from scipy.stats import spearmanr
ModuleNotFoundError: No module named 'scipy'
PS C:\Users\sodiq\OneDrive\Desktop\Crypto-scanner> python backtest_v2.py --from .\backtest_v2_20260928_1446.csv
collecting data for 328 coins x 16 test dates (01 Jun to 14 Sep)...
```

```
PS C:\Users\sodiq\OneDrive\Desktop\Crypto-scanner> python backtest_v2.py --from .\backtest_v2_20260928_1446.csv
At line:1 char:2
+ ~ python backtest_v2.py --from .\backtest_v2_20260928_1446.csv
+ ~
Missing expression after unary operator '-'.
At line:1 char:5
+ ~ python backtest_v2.py --from .\backtest_v2_20260928_1446.csv
+ ~~~~~
Unexpected token 'python' in expression or statement.
+ CategoryInfo          : ParserError: (:) [], ParentContainsErrorRecordException
+ FullyQualifiedErrorId : MissingExpressionAfterOperator
```

Lesson: save the raw data first, so a crash never costs you the whole run.

The fair test: 961 samples, 16 weeks

Every coin compared to the median coin on the same date, so a market rally can't fake good results.

```
PS C:\Users\sodiq\OneDrive\Desktop\Crypto-scanner> python backtest_v2.py --from .\back
```

```
BACKTEST v2 REPORT - 2026-09-28 15:09
```

```
Samples: 961 | coins: 132 | test dates: 16
```

```
Excess = return vs the median coin on the same date (removes market-wide rallies).
```

```
=== 3-DAY HORIZON (hit = touched +15%) ===
```

bucket	n	avg excess	median exc	beat mkt	hit rate	avg dip
<30	753	-0.4%	+0.0%	50%	8%	-7.4%
30-49	163	+0.9%	-0.1%	48%	10%	-6.6%
50-64	36	+0.2%	-0.0%	50%	6%	-8.0%
65-79	8	-1.0%	-1.3%	38%	12%	-9.5%
80+	1	+10.2%	+10.2%	100%	0%	-2.3%
ALL	961	-0.1%	+0.0%	50%	8%	-7.3%

```
Score vs excess correlation: +0.019 (>+0.05 = useful, ~0 = no edge, <0 = backwards)
```

Correlation: 0.02 to 0.10. Almost no edge.

Most of my "smart" signals hurt

Extra 14-day return when each signal was present vs. absent:

Beaten down + hasn't moved yet (filters)	+7.1%
Long, tight price base	+2.1%
Volatility squeeze	+1.0%
Trend flip	-0.9%
Volume waking up	-1.3%
Buy-volume dominance	-2.2%
OBV accumulation	-3.0%
Strength vs Bitcoin	-3.2%

Momentum signals fire AFTER the move starts. The edge is in the quiet coil before it.

Rebuilt around the evidence

Coil = the core score. Momentum = entry timing only.
Plus free news feeds for catalysts, events and red flags
(hacks, delistings). 100% free data.

```
9  ● TRIGGER coiled + (catalyst / hot sector / squeeze fuel) + momentum just confirmed -> entry zone
10 ● HEATING coiled + (catalyst / hot sector / squeeze fuel) -> get ready
11 ● WATCH  coiled, nothing else yet -> keep an eye on it
12 ● RED FLAG hack / exploit / delisting / monitoring-tag news -> avoid
13
14 USAGE
15 python scanner_v3.py -> scan once
16 python scanner_v3.py --every 4 -> keep scanning every 4 hours (VS Code must stay open)
17 python scanner_v3.py --install-task 4 -> AUTOMATIC: Windows runs it every 4h in the background
18                                     (use 1 for hourly, 24 for daily at 08:00)
19 python scanner_v3.py --remove-task -> stop the automatic scans
20 python scanner_v3.py --sniper -> watch Bybit/Binance listing announcements every 5 min
21 python scanner_v3.py --coin GRAM -> detailed breakdown for one coin
22 python scanner_v3.py --stats -> how past alerts actually performed (forward test)
```

● **TRIGGER** coiled + catalyst + momentum confirmed

● **HEATING** coiled + news / hot sector / squeeze fuel

● **WATCH** coiled, nothing else yet

● **RED FLAG** hack / exploit / delisting news: skip

Runs every 4 hours. Results land on my phone.

Every alert is logged and checked 3, 7 and 14 days later.
That's the live forward test.

🔍 CRYPTO SCAN 28 Sep 15:48
BTC: 🟢 BULL (above 200 EMA)
🟢 Trigger 2 🟡 Heating 3 🟡 Watch 4 🔴 Flags 5

🏆 TOP 9 COINS

🏆 🟢 TRIGGER XDC \$0.03393 score 75
✅ Coil 🟡 News ✅ Sector 🟡 Fuel ✅ Trigger
• 120d base + tight squeeze
• 🔥 hot sector: Real World Assets (RWA) +49.3%
• 📺 trigger ON (vol 4.9x, +7% vs 200 EMA)

🏆 🟢 TRIGGER GRAM \$1.617 score 70
✅ Coil 🟡 News 🟡 Sector ✅ Fuel ✅ Trigger
• 90d base + tight squeeze
• ⚡ open interest +24% in 7d
• 📺 trigger ON (vol 3.0x, +3% vs 200 EMA)

🏆 🟡 HEATING XLM \$0.2172 score 52
✅ Coil 🟡 News ✅ Sector ✅ Fuel 🟡 Trigger
• 90d base + squeezing
• 🔥 hot sector: Real World Assets (RWA) +49.3%
• ⚡ open interest +19% in 7d

4. 🟡 HEATING ASTER \$0.6952 score 40
✅ Coil ✅ News 🟡 Sector ✅ Fuel 🟡 Trigger
• 120d base
• 📺 Aster Launches Perpetual Grid Trading 2.0 with Up to 140,000 \$ASTER Li
• ⚡ negative funding -0.005%

5. 🟡 HEATING JTO \$0.5554 score 35
✅ Coil 🟡 News 🟡 Sector ✅ Fuel 🟡 Trigger
• volatility squeeze
• ⚡ negative funding -0.001%

What I learned

- 1** A backtest that proves you wrong beats a strategy that just looks good.
- 2** Check your test for bias before trusting the result.
- 3** Simple, evidence-backed signals beat 13 clever ones.
- 4** Save raw data first. Crashes happen.
- 5** AI is a great coding partner. The thinking, testing and judgment are still yours.

Now forward-testing live. I'll share the results.

Not financial advice.